

Deploy and Test a GKE Autopilot Workload

120 minutes · Mapped criteria A3, A4, A5

Purpose and work product

Deploy a containerized web workload with probes and resource requests, add HPA and diagnose why replicas do not scale.

Submit: Kubernetes manifests, load-test CSV and HPA evidence

Safety, cost and privacy

Use only the assigned training project. Confirm the active account, project and region before every mutation. Never place credentials, payment details, access tokens or private keys in screenshots or submitted files. Stop billable resources and verify the final inventory.

Files in this folder

File	Purpose
deployment.yaml	Mock data, script or configuration used by this activity
hpa.yaml	Mock data, script or configuration used by this activity

Detailed procedure

Step 1

Set the project and region, enable the GKE API and create ace-gke in Autopilot mode.

Step 2

Fetch credentials and run `kubectl config current-context`; confirm the project, cluster and region.

Step 3

Review `deployment.yaml` for image, labels, ports, resource requests/limits and readiness probe.

Step 4

Apply `deployment.yaml` and wait for both replicas to become Ready.

Step 5

Apply `hpa.yaml`. Run `kubectl get hpa` and confirm the TARGETS column has metrics rather than unknown.

Step 6

Generate controlled HTTP load from an approved load generator for at least five minutes.

Step 7

In a second terminal, record HPA current/target CPU, desired replicas and Pod count every 30 seconds.

Step 8

If replicas remain unchanged, check CPU requests, Metrics API output, workload intensity, `maxReplicas` and recent events.

Step 9

Correct exactly one identified cause and repeat the same load profile.

Step 10

Save `kubectl describe hpa`, events, Pod status and the load-test CSV.

Step 11

Scale the workload down or delete the cluster after evidence is captured to avoid ongoing cost.

Verification record

Record expected and observed output. A successful command without the correct project, identity, scope or evidence is not an accepted result.

Check	Expected evidence
Identity and scope	Active account, project ID, region or zone
Configuration	Resource description or policy output
Runtime result	Test output with UTC timestamp
Failure test	One controlled negative test and diagnosis
Cleanup	Final inventory and stopped/deleted billable resources

Troubleshooting sequence

Check the active identity and project first, then API enablement and IAM, then resource state, networking and quotas. Preserve the failed output before making a correction. If the issue remains, escalate with the exact command, timestamp, expected result, observed result and evidence files.

Cleanup

Delete or stop resources as instructed. Re-open Billing and the relevant service inventory, then record the remaining resource names. Budget alerts do not stop charges.